A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 06

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Light:** selective felling to create clearings for plants needing high light while retaining dense canopy for shade species – clearings also give warm sand / soil for lizard eggs – leave dead wood – woodland floor plants (bluebells) grow before the canopy leafs **2 Dissolved oxygen:** narrowing a slow river increases flow and turbulence so more oxygen dissolves, suiting trout and mayfly larvae – shallow water warms quickly for egg development **2 Nutrients:** removing fertile topsoil / not fertilising allows plants of nutrient-poor soils (eg sundew on acidic nitrate-deficient soils) that cannot compete on fertile ground – dung patches add local nutrients **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **biodiversity** → the variety of life: habitat, species and genetic diversity	6

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Food:** wildflower grasslands for seed-eating birds – providing food species **1 Control of predation:** island nest sites unreachable by foxes – trapping / culling / eradicating introduced rats (Scilly Isles - Manx shearwater and storm petrel breeding success increased) **1 Control of competitors:** removing rhododendron so wildflowers get light **1 Pollination:** a range of flowering plants with different flowering times so pollinators have food all season **1** Seed dispersal species (hippos, elephants, monkeys) conserved – control of introduced pathogens **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 43-46; AQA spec 3.1.2.3.4, 3.1.3.1.	
02.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Designation of protected areas (SSSIs, MCZs) **1** Protection of wild birds and their nests (except some pests / game birds) **1** Protection of named mammals such as otters, dormice, red squirrels, badgers and their setts **1** Uprooting wild plants generally illegal **1** Bats must not be disturbed – roof timber treatments must be non-toxic to bats **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 4 points below. **1 Appendix I:** species threatened with extinction - all international (commercial) trade banned, except movement for conservation breeding **1 Examples:** great apes, big cats, rhinos, blue whale **1 Appendix II:** species that may become threatened if trade is not closely controlled - trade permitted under licence from countries where the species is well protected **1 Examples:** Honduras mahogany, hippopotamus, green iguana, great white / basking shark, Venus fly trap **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **CITES** → the treaty that controls international trade in endangered species **Do not credit** • 'CITES bans all trade in wildlife' - it controls and licenses trade, with different rules for each appendix	4
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Collection may be hard to police in remote areas / poor countries, but if the animals or products cannot be exported / imported there is no market **1** Removing the profit removes the incentive to collect **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30-31; AQA spec 3.1.2.3.2.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Plants close their stomata to reduce transpiration losses **1** Closed stomata stop the absorption of carbon dioxide so photosynthesis and growth stop **1** Less water for nutrient absorption / transport and cell turgidity – severe shortage dehydrates cells and inhibits biochemical reactions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves	3

Do not write outside the box

Question	Marking guidance	Marks
03.2	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2** Anaerobic soil favours denitrifying bacteria and inhibits nitrifying bacteria so nitrate is lost faster than it forms – higher risk of fungal disease – root respiration inhibited **up to 2** Drainage ditches or pipes **up to 2** Deep ploughing – avoiding compaction by machinery / livestock – adding organic matter to encourage worms **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **waterlogged** → soil pores filled with water so there is no air • **denitrifying** → nitrate converted back to nitrogen gas • **nitrifying** → ammonium oxidised to nitrite and then nitrate	4
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Groundwater often contains dissolved salts – when the water evaporates from the soil the salts are left behind and accumulate **1** High soil salinity causes osmotic dehydration of roots / crop death – washing out with extra water uses more water and salinises rivers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **salinisation** → salt building up in the soil / soil becoming salty	2
03.4	1 mark for the point below. **1** Add crushed lime (calcium carbonate / hydroxide) – low pH increases leaching, inhibits nutrient uptake and mobilises toxic aluminium and lead **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leaching** → nutrients or ions washed down out of the soil / washed away by water	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 308-310, 313; AQA spec 3.5.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 6 points below. **1** Cycling of materials (like biogeochemical cycles) - recycling, re-use **1** Energy from renewable sources **1** Human activities should support ecosystems / conserve biodiversity **1** Separation of technical and biological materials – connected systems where the waste of one process feeds another (industrial symbiosis) **1** Diverse systems are more resistant to change **1** Design of products for extended use and end-of-life re-use – optimum rather than maximum production – technologies that improve efficiency and enable dismantling **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new • **biodiversity** → the variety of life: habitat, species and genetic diversity • **symbiosis** → a close relationship in which both species benefit **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	6
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Natural systems over-produce so the surplus supports the processes the system relies on: plant products feed pollinators, seed dispersers, decomposers, mycorrhizae **1** Agroecosystems aim for maximum harvested yield, removing the surplus that would maintain soil biota, pollinators and pest predators, so they need artificial inputs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Costs of environmental measures could put companies in one country at a disadvantage against competitors in countries without them **1** A common approach gives a level playing field, coordinated monitoring and support for less successful countries **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 374-375; AQA spec 3.6.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 4 points below. **1** Natural cycles use a small number of abundant elements (C, H, O, N, P) built into monomers and polymers (carbohydrates, proteins) – wastes of one process are raw materials for another so nothing is depleted or accumulates **1** Natural wastes are biodegradable / non-toxic and are broken down by decomposers into raw materials **1 Human systems are linear:** extract - process - manufacture - use - discard – minerals are dispersed after use or mixed so recovery is difficult **1** Human wastes are often persistent and toxic (organochlorines, PCBs, heavy metals in electronics) and build up – fossil fuel use is inherently linear and unsustainable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Metals are dispersed in small amounts within complex mixed products so they cannot easily be separated / recovered at end of life **1** They leach from landfill or are released by incineration – persistent, bioaccumulating **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainability** → meeting present needs without stopping future generations meeting theirs • **incineration** → burning waste, usually with energy recovery • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
05.3	**1 mark each** 1 mark for each of the 4 points below. **1** Design for durability, repair and re-use (replaceable parts, no built-in batteries, spare parts and support, no planned obsolescence) **1** Easy identification of materials (coding) and separation of components so they can be recycled **1** Separation of technical materials (metals, plastics) from biological materials (which can be composted / digested) **1** Use of recycled and recyclable / biodegradable materials – reduced packaging and material use (Cradle to Cradle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers	4

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.3.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B	1
06.2	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Harvesting no faster than the MSY - resource never depleted **1** Mixed species plantations - more food choices, relationships and niches **1** Indigenous species - support indigenous insects, birds **1** Mixed age structure - varied light, temperature, wind so more species – continuous cover and yield **1** Selective logging - small clearings, less disturbance, long-term timber benefit **1** Retention of dead wood, buffer zones, corridors **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **MSY** → the largest catch that can be taken without reducing the stock **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369; AQA spec 3.5.3.	
07.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Sinkers / weighted lines so hooks sink quickly below the reach of albatrosses **1** Setting lines at night when birds are not feeding **1** Bird-scaring streamers / decoys **1** Curved hooks that catch tuna but not birds **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 3 points below. **1 Effort:** limits on boat size, engine power, days at sea, number of licences – decommissioning payments to reduce fleet size where overfishing is unavoidable **1 Methods:** bans or length limits on drift nets – bans on demersal trawling over sensitive seabed (Rockall Bank, Hatton Bank deep-water reefs) **1** Shorter trawl tows so turtles / by-catch survive (over 99% survive under 10 minutes) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **decommissioning** → safely dismantling a power station at the end of its life **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	3
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Eggs / larvae are reared in captivity through the planktonic stage when most die (predation, failure to find habitat) **1** Juveniles are released when survival chances are good, increasing recruitment to the adult population **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Egg / larval survival varies greatly between years so adult numbers do not predict recruitment **1** Several poor years would lower the MSY – egg surveys allow catches to be reduced in advance rather than after a sudden collapse **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **MSY** → the largest catch that can be taken without reducing the stock	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.	
08.1	**1 mark each** 1 mark for each of the 2 points below. **1** Giant tritons are predators of the crown-of-thorns starfish which eats coral polyps **1** Removing tritons allowed starfish populations to explode, destroying large areas of reef (food web / trophic cascade) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	2

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 4 points below. **1** Fixed mooring buoys so boats do not anchor on coral **1** Education of divers / snorkellers not to touch coral or kick up sediment – bans on spear fishing and souvenir collection **1 Fishing controls:** no-take zones (Chagos, Great Barrier Reef, Lundy, Lamlash Bay), seasons, minimum and maximum catch sizes, bans on dynamite / small-mesh nets / trawling **1** Protection of turtle nesting beaches – control of coastal development – protection of mangroves and seagrass – artificial reefs (reef balls, sunk ships / oil rigs) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** No sunlight so no symbiotic algae – polyps rely only on captured food so have far less energy and grow very slowly **1** Damage from trawls takes hundreds of years to repair – the slow-growing, low-fecundity fish (orange roughy) they support are easily over-exploited **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **symbiotic** → a close relationship in which both species benefit	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Deep-water trawling (also by-catch) – oil and gas exploration in deeper water – ocean acidification **1** Protected areas banning trawling, eg Darwin Mounds SAC off NW Scotland, Oculina Bank off Florida **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 61-64; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2** Drainage of wetlands / lowering water tables by groundwater abstraction dries out wetlands – HEP schemes alter river levels flooding sandbank nests of turtles / birds – species with narrow tolerance (specialists) affected first – sundews lost when drained soil becomes aerobic / nitrogen-rich and taller competitors invade **2** Hot water discharges and organic wastes (sewage) reduce dissolved oxygen, killing aerobic organisms such as fish and insect larvae **2** Acid mine drainage and acid rain gases lower pH, denaturing proteins of exposed tissues (fish eggs, gills) and dissolving calcium exoskeletons (crayfish) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **abstraction** → water taken out of a river, lake or aquifer • **HEP** → electricity generated from falling water **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	6
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Reduces light penetration so submerged plants cannot photosynthesise **1** Blocks the gills of filter feeders such as bivalve molluscs – sediment smothers the bed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Rare species often have very specialised habitat requirements / narrow ranges of tolerance **1** A small change moves conditions outside their range while generalists survive – small populations cannot recover **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **tolerance** → the range of conditions in which a species can survive	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 22-23; AQA spec 3.1.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Succession:** pioneers, seres (hydrosere, lithosere, psammosere), modification of abiotic conditions, climax communities controlled by climate – deflected succession and plagioclimaxes • **Application:** deciding whether to manage (plagioclimax habitats need grazing, mowing, burning, coppicing, pollarding continued) or leave alone (climax habitats) – habitat creation planning the stages – controlling scrub / secondary succession – understanding that new habitats change over time • **Population dynamics:** birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles, r- and k-selection, MSY • **Application:** setting sustainable harvest levels (fisheries, whaling, forestry) – recognising k-selected species need strict protection and recover slowly – deciding when culling is needed (deer without wolves, introduced species) or when releases / captive breeding are needed (low breeding rate) – predicting effects of reintroducing predators • Monitoring population size, distribution, survival rate and age structure to assess success – adaptation and range of tolerance guiding abiotic management • **Evaluation:** uncertainties (natural fluctuations, unknown carrying capacity) – examples (Yellowstone wolves, Scottish deer, chalk grassland) – judgement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• **succession** → the gradual change in the community over time • **abiotic** → non-living / physical factors • **deflected succession** → a community held back from its climax by human activity **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	

Do not write outside the box

Question	Marking guidance	Marks
	Model answer - Level 3 (8 marks)	
	Succession is the sequence by which pioneers colonise bare ground, modify the abiotic conditions and are replaced by later seres until a climax community controlled by climate is reached; where grazing, mowing, burning or coppicing halts it, a plagioclimax forms. This tells a conservation manager whether to intervene: chalk grassland, heathland and coppiced woodland are plagioclimaxes that disappear without continued management, whereas a climax habitat is best left alone, and habitat creation must plan the stages, control scrub and manage secondary succession.	
	Population dynamics - birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles and r- and k-selection - inform decisions of a different kind. Maximum sustainable yield sets harvest levels in fisheries, whaling and forestry; k-selected species that breed late and slowly need strict protection and recover slowly; culling may be justified where predators are absent, as with Scottish red deer without wolves, or where introduced species threaten natives, while the reintroduction of wolves to Yellowstone shows how restoring a predator changes vegetation through the food web. Monitoring population size, distribution, survival and age structure measures success, and knowledge of adaptation and ranges of tolerance guides how abiotic conditions are managed.	
	The limits are uncertainty: natural fluctuations, unknown carrying capacities and the difficulty of predicting how communities respond. Ecological understanding cannot remove that uncertainty, but without it conservation decisions - to manage or not, to cull or protect, how much to harvest - are guesses.	
	Weaker answer - Level 2 (5 marks)	
	Succession is when a habitat changes over time from pioneer species to a climax community. Conservationists need to understand this because some habitats like heathland will turn into woodland if they are not managed by grazing or burning. Population dynamics is about how populations grow and are limited by factors like food and predators. This helps conservation because you can work out how many animals can be taken without the population declining, and when a population needs protecting. Monitoring populations shows if conservation is working.	
	What makes the difference	
	• Level 2 explains both concepts correctly and links them to decisions, but generally; Level 3 uses the technical vocabulary (seres, plagioclimax, k-selection, MSY) with named cases (Yellowstone wolves, Scottish deer, chalk grassland).	
	• Level 3 covers more applications (habitat creation, culling introduced species, adaptation and tolerance) and evaluates the uncertainties.	
	• Structure: Level 3 gives each concept its own paragraph with the decisions it informs, then a conclusion.	

Do not write outside the box

Question	Marking guidance	Marks
10.2	1 mark for the point below. **1** Any taxonomic group of organisms classified together by similarity (eg species, genus, family) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-85; AQA spec 3.1.3.	
11.1	**Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Energy flow:** autotrophs and heterotrophs – losses by reflection, transmission, respiration – 100 000 to 1 000 units – few food chains beyond four levels – plant food most productive – grazing animals where crops impossible – pigs on wastes • **Energetics:** intensive vs extensive, law of diminishing returns, energy subsidies (fertiliser, pesticides, irrigation, machinery, drying, processing, transport), energy ratios, food conversion ratios, dependence on fossil fuels • **Control of genetics:** asexual reproduction / cuttings / cloning, selective breeding and breeds, cross-breeding and hybrid vigour, AI and embryo transfer • **GM:** Roundup Ready soya, Bt crops, golden rice, omega-3 oilseed rape – advantages – risks (gene transfer, antibiotic markers, allergies, costs to LEDC farmers, loss of local diversity)	25

Do not write outside the box

Question	Marking guidance	Marks
	• **Sustainability links:** energy dependence, trophic level of diet, feeding 12 billion people • Justified conclusion on the balance between productivity, efficiency and sustainability *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	
11.2	**Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Ecological features:** cnidarian polyps, cilia, nematocysts, zooxanthellae symbiosis, limestone skeletons building reefs, complex structure and food web – specific abiotic requirements (temperature, light, salinity, low turbidity) • **Importance:** quarter of marine species, fisheries, medicines, carbon sink, erosion protection, tourism • **Vulnerability:** narrow abiotic tolerance means sensitivity to warming, acidification, sedimentation, nutrient pollution, oil, bleaching – slow growth so slow recovery – physical damage – overfishing and food-web effects (crown of thorns) – introduced species – dependence on mangroves and seagrass • **Conservation:** MPAs, No Take Zones, fishing controls, tourism management, protection of linked habitats, CITES • Comparison with other habitats (rainforests, mangroves, Antarctica) where relevant • Evaluation of local vs global threats and justified conclusion	25

Do not write outside the box

Question	Marking guidance	Marks
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	

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